COURSE DESCRIPTION FORM: EE-1005 DIGITAL LOGIC & DESIGN

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM(S) TO BE EVALUATED [BS-CS] **Spring 2024**

Course Description

Course Code	EE-1005
Course Title	DIGITAL LOGIC DESIGN (CS)
Credit Hours	4 (3+1)
Prerequisites by Course(s) and Topics	None
Grading Policy	Absolute Grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam retake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake cases.
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	*100% Theory [and a separate 1 credit hour lab]* Assessment items of Theory Part (see table below)
Course Instructors	Dr. Mehwish Hassan, Mr. Shams Farooq, Mr. Amir Gulzar,
Lab Instructors (if any)	Ms. Ifrah Maqsood, Mr. Sohail Aziz
Course Coordinator	Mr. Shams Farooq
URL (if any)	
Current Catalog Description	Introduction to Number Systems; Binary Logic and Gates; Combinational Circuits; Sequential Circuits; Registers and Counters; Programmable Logic Technologies and Emerging Trends
Textbook (or **Laboratory Manual**	**Digital Design,** M. Morris Mano, Michael Ciletti, Pearson, 5^{th} Edition

Assessment Item	**Number**	**Weight (%)**
Assignments	5(combined- all sections)	10
Quizzes	6-10 (can vary)	20
Sessional-I	1 (combined- all sections)	15
Sessional-II	1 (combined- all sections)	15
Final Exam	1 (combined- all sections)	40

for Laboratory Courses)	
Reference Material	**Digital Fundamentals,** Thomas L. Floyd, PEARSON, 10th Edition **Logic and Computer Design Fundamentals by** M. Morris Mano, Charles Kime.
Course Learning Outcomes	**A. Course Learning Outcomes (CLOs)** The course objective is to prepare students to develop understating for the application of Digital Logic Design concepts to the current systems with hands-on-experience in a team work environment. After completion of the course, the student shall be able to: 1. Develop an understanding of the foundations about the logic of computer design. 2. Understand different number systems and code along with their inter-conversions and signed implementations. 3. Apply Boolean expression onto circuit design and minimize such expressions using k-maps. 4. Analyze and design combinational circuits such as decoders, encoders, priority encoders, multiplexers, binary adders (half, full and carry look ahead) and binary subtractors. 5. Analyze and design sequential circuits, i.e. latches (SR, SR with control input, D) and flip flops (D, T and J/K). 6. Understand and show variations of counter and registers. 7. Develop an understanding for memory and programmable logic devices. Connect the concepts learnt in class rooms with practical implementation in laboratory.

	B. Program Learning Outcomes	
	For each attribute below, indicate whether this attribute is covered in this c or not. Leave the cell blank if the enablement is little or non-existent.	
1. Computing Knowledge:	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems	✔
2. Problem Analysis:	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.	✔
3.Design/ Develop Solutions:	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✔
4. Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods.	✔

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5. Modern Tool Usage:	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✔	
6. Society Responsibility:	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.		✔
7. Environment and Sustainability	Understand and evaluate sustainability and impact of professional computing work in the solution of complex computing problems.		
8. Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.		
9.Individual and Teamwork:	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.	✔	
10. Communication	Communicate effectively on complex computing activities with the computing community and with society at large	✔	
11. Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member of a team.	✔	
12. Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.	✔	

C. Mapping of CLOs on PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs											
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CLOs	1	✔					✔				✔		
	2	✔			✔					✔	✔		
	3	✔	✔		✔		✔			✔	✔		✔
	4	✔	✔	✔	✔					✔	✔		✔
	5	✔	✔	✔	✔					✔	✔	✔	✔

	6	✔	✔	✔	✔	✔				✔	✔	✔	
	7.	✔				✔				✔	✔	✔	

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)

Topics to be covered:

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction, Number Systems (Binary, Octal and hexadecimal), Number Ranges, Arithmetic Operations, Conversion from Decimal to Other Bases, Negative number representations – Chapter 1	**2**	**6**	**1,2**
Binary Logic and Gates, Boolean Algebra, Standard Forms, Map Simplification (SOP & POS), Map Manipulation, Don't-Care Conditions, NAND, NOR & Exclusive-OR Gates & Circuits, Integrated Circuits, Positive & Negative Logic – Chapter 2 & 3	**4**	**12**	**3**
Sessional-I	**0.5**	**1**	
Combinational Circuits, Analysis Procedure, Design Procedure, Decoders, Encoders, Priority Encoders, Multiplexers, Demultiplexers, Binary Adders (Half, Full, Ripple Carry, Carry Lookahead), Binary Subtraction, Signed Binary Numbers, Overflow – Chapter 4	**2**	**6**	**4**
Sessional-II	**0.5**	**1**	
Sequential Circuits, Latches, Flip-Flops, Sequential Circuit Analysis, State Diagram, Sequential Circuit Design (with D Flip-Flops, JK Flip-Flops, T flip flops), word problems – Chapter 5	**2**	**6**	**5**
Registers and Counters, Register with Parallel Load, Shift Registers, Shift Register with Parallel Load, Bidirectional Shift Register, Ripple Counter, Synchronous Binary Counters, Serial and Parallel Counters, Up-Down Binary Counter, Binary Counter with Parallel Load – Chapter 6	**2**	**6**	**6**

	Programmable Logic Technologies (Read-only Memory, Programmable Logic Arrays, Programmable Array Logic Devices, introduction to FPGAs) – Chapter 7	**2**	**6**	**7**
	Revision	**1**	**3**	
	Total	**16**	**45**	

Laboratory Projects/Experiments Done in the Course	Introduction to Trainer Board and Integrated Circuits. Hands-on experience with basic logic gates and other logical components used in embedded systems. Understanding of Datasheets of ICs. Debugging of ICs and combinational circuits.
Programming Assignments Done in the Course	None

Class Time Spent (in hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	40	25	30	5

Oral and Written Communications	Every student is required to submit at least __4___ written reports of typically _5____ pages

Lab/ Practical Component of the course

[Please remove this part if there is no Lab component associated with this course]

[Example of DB course is added to provide template]

Weeks	Contents/Topics	Assessment Items (Case Study/ Exercise Assignment/ Quiz etc.)
Week-01	Introduction to the Trainer Board and Integrated Circuits	
Week-02	Logic Gates(AND, OR, NOT, XOR, XNOR, NAND, NOR)	
Week-03	Proteus	Lab Assignment
Week-04	NAND & NOR Universal Gates	Lab Quiz-1
Week-05	Boolean Simplification	
Week-06	Half Adder and Full Subtractor	Case Study
Week-07	Magnitude Comparator	Lab Assignment
Week-08	Decoder & Priority Decoder	
Week-09	Sessional Exam	Lab Assignment
Week-10	BCD decoder to Seven segment LED display	Lab Quiz-2
Week-11	8-to-1 and 4-to-1 Multiplexer	
Week -12	555 Timer	
Week 13	Latches and Flip Flops	
Week-14	4-bit register which can shift left, right and load data in parallel	
Week-15	Counters	

Practical/ Programming Work/ Tools: Trainer Board- Proteus